

**EXP NO: 1**

## **AZURE DEVOPS ENVIRONMENT SETUP**

### **Aim:**

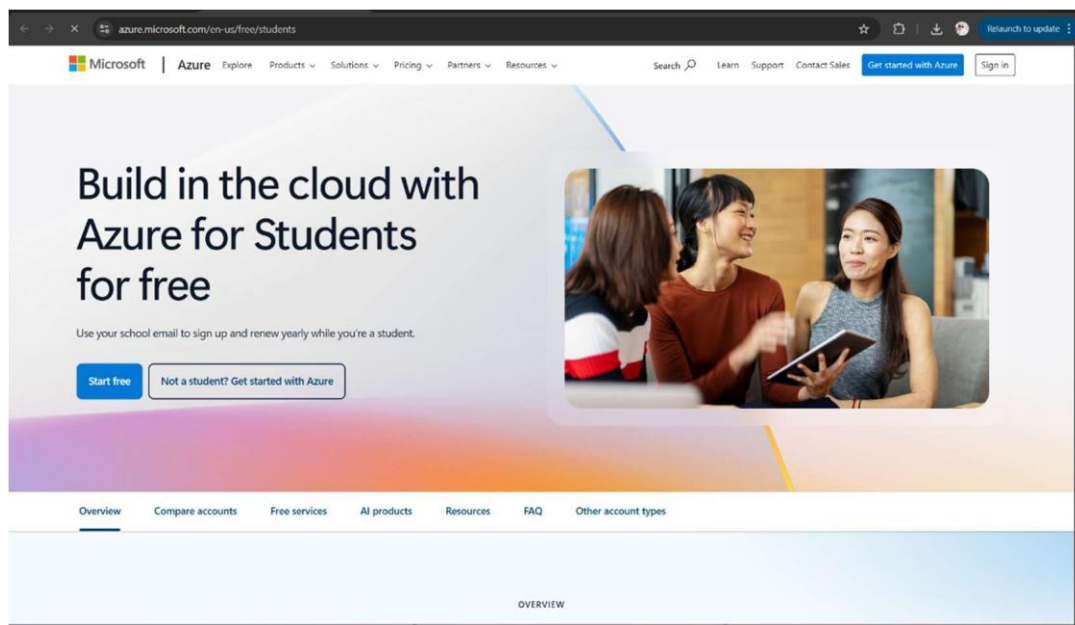
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

### **INSTALLATION**

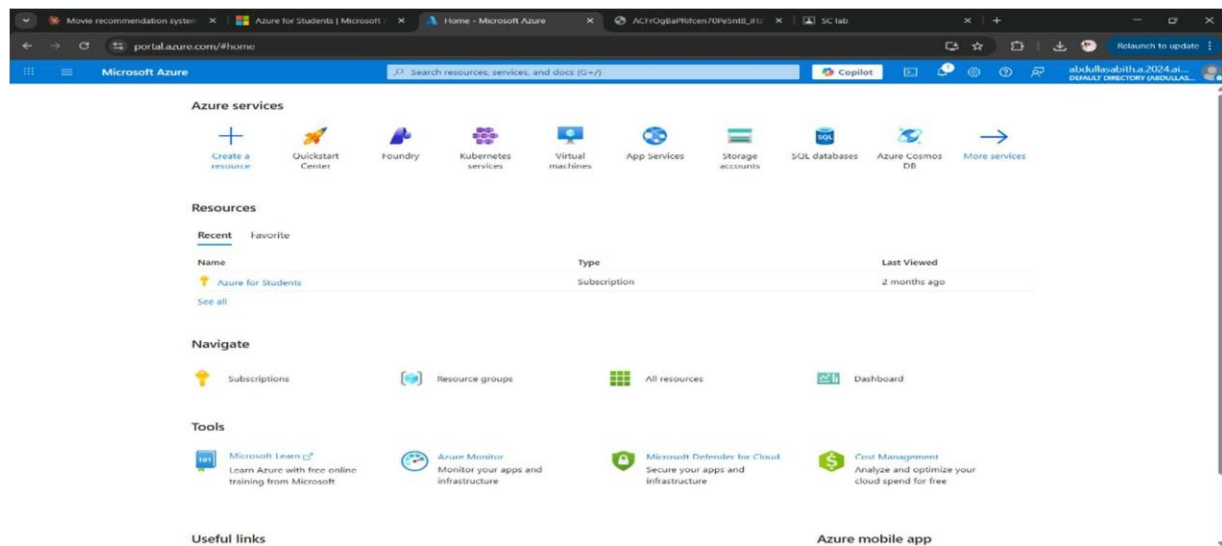
1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/getstarted/azureportal>.

Sign in using your Microsoft account credentials.

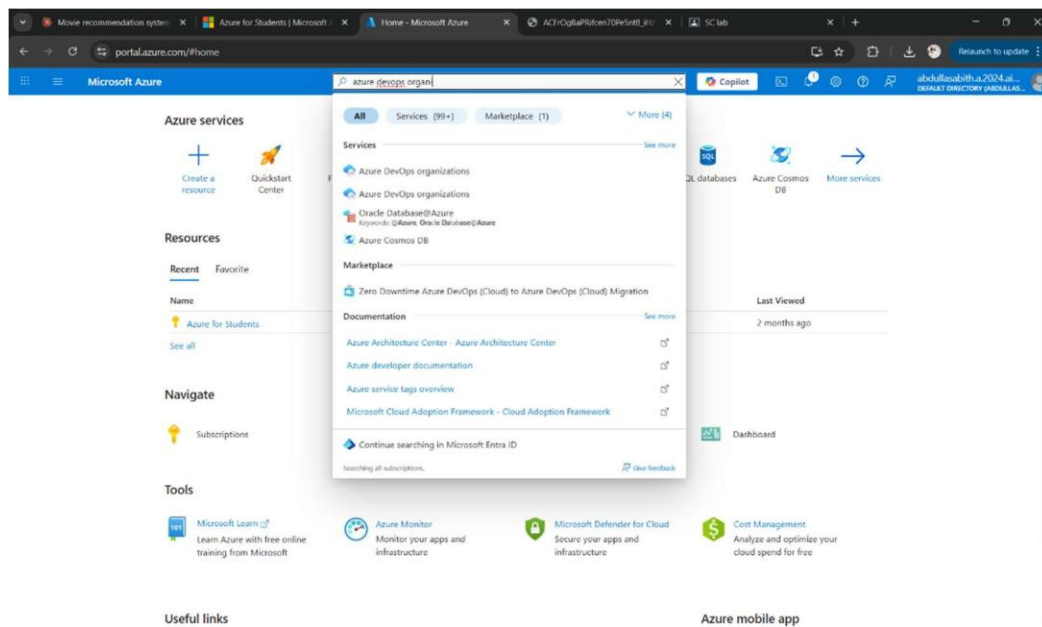
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



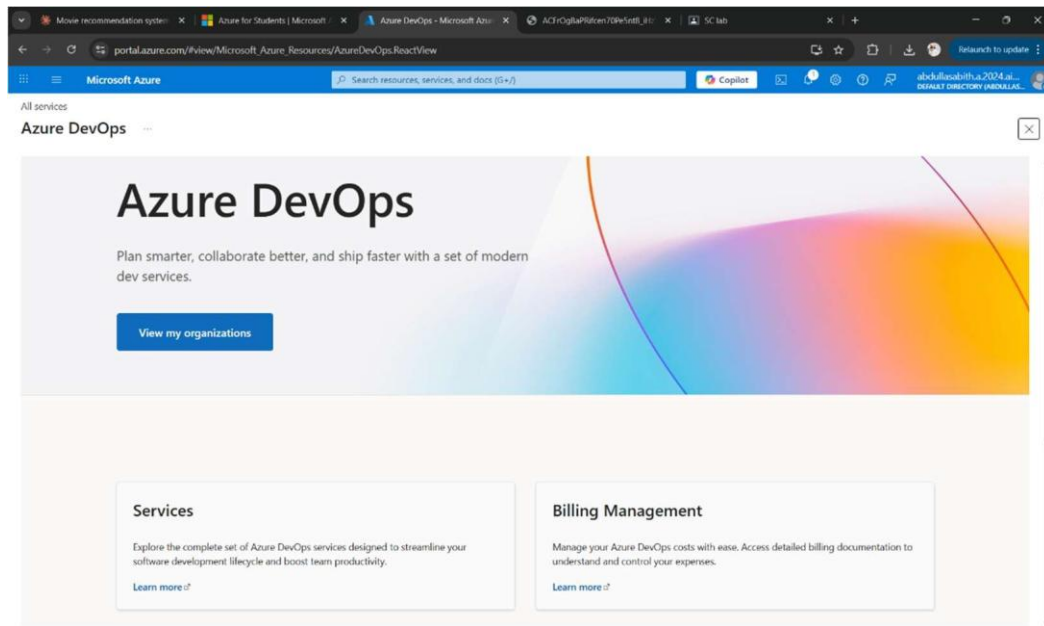
2. Azure home page



3. Open DevOps environment in the Azure platform by typing Azure DevOps Organizations in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



**Result:**

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2

## AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

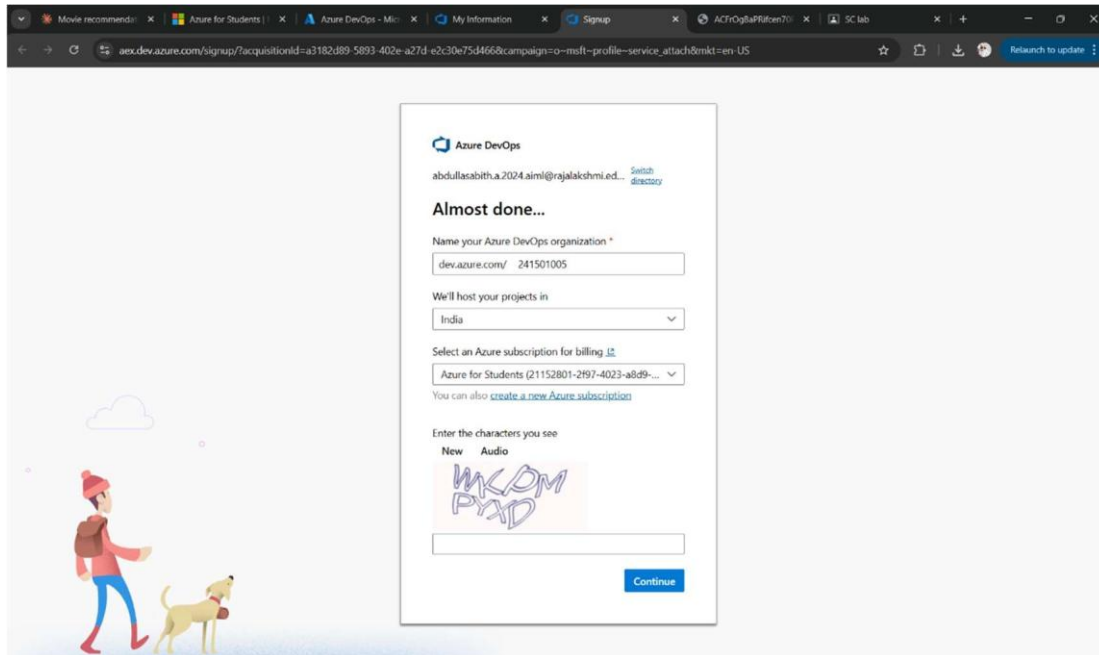
### Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

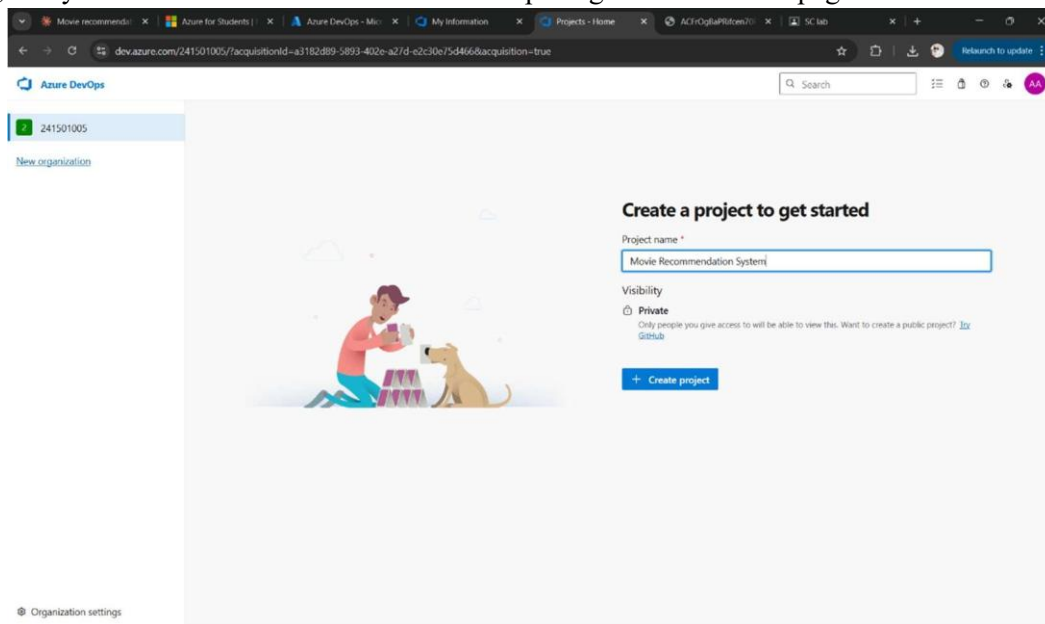
### 1.Create An Azure Account

### 2.Create the First Project in Your Organization

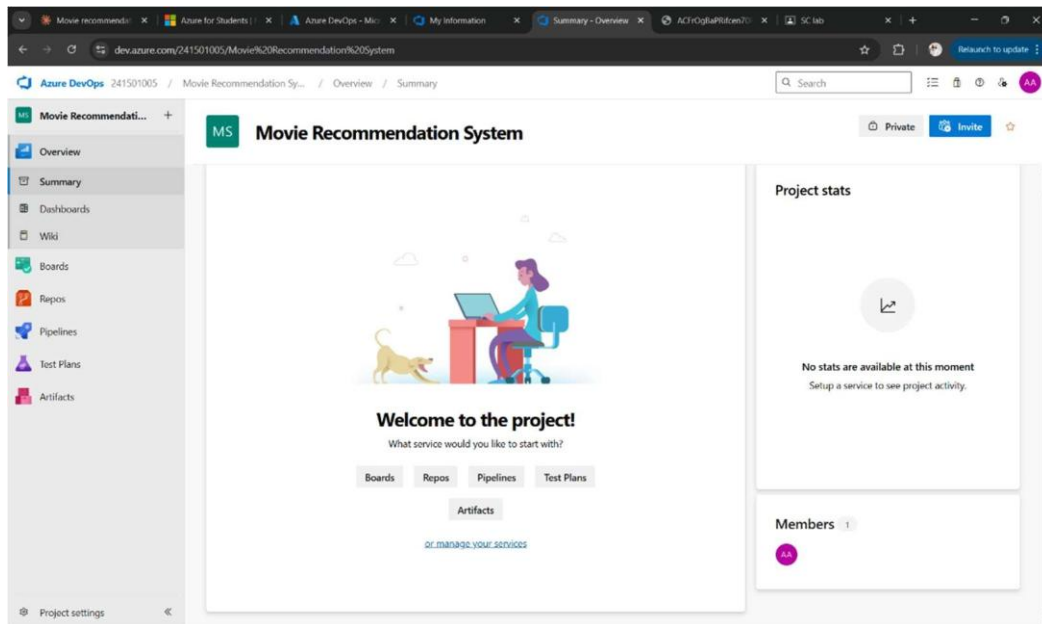
- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
  - Name:** Choose a name for the project (e.g., **LMS**).
  - Description:** Optionally, add a description to provide more context about the project.
  - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.



3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



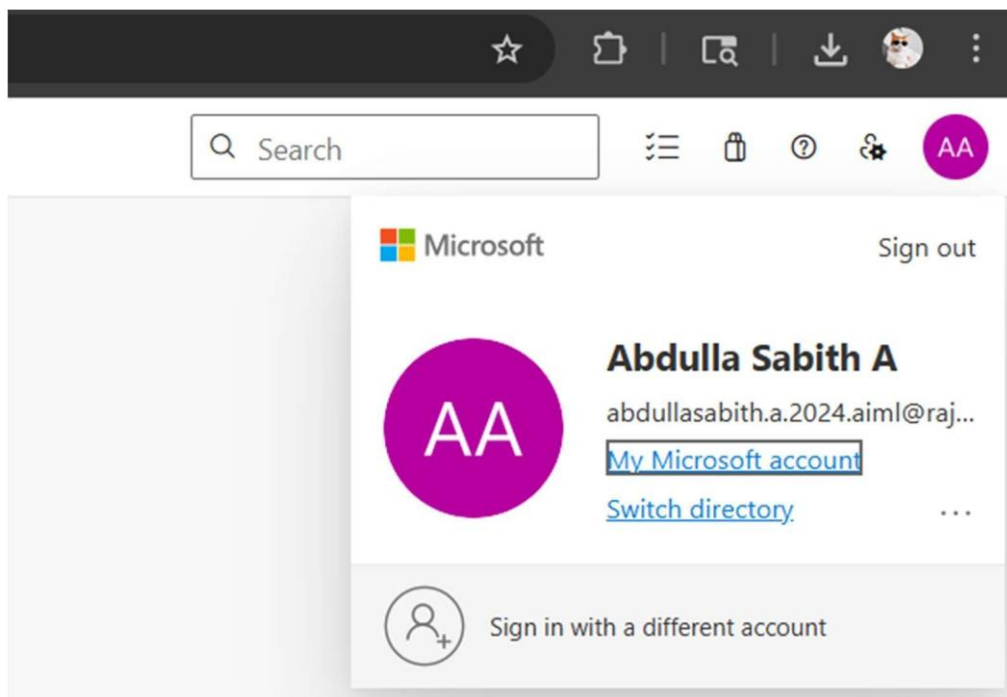
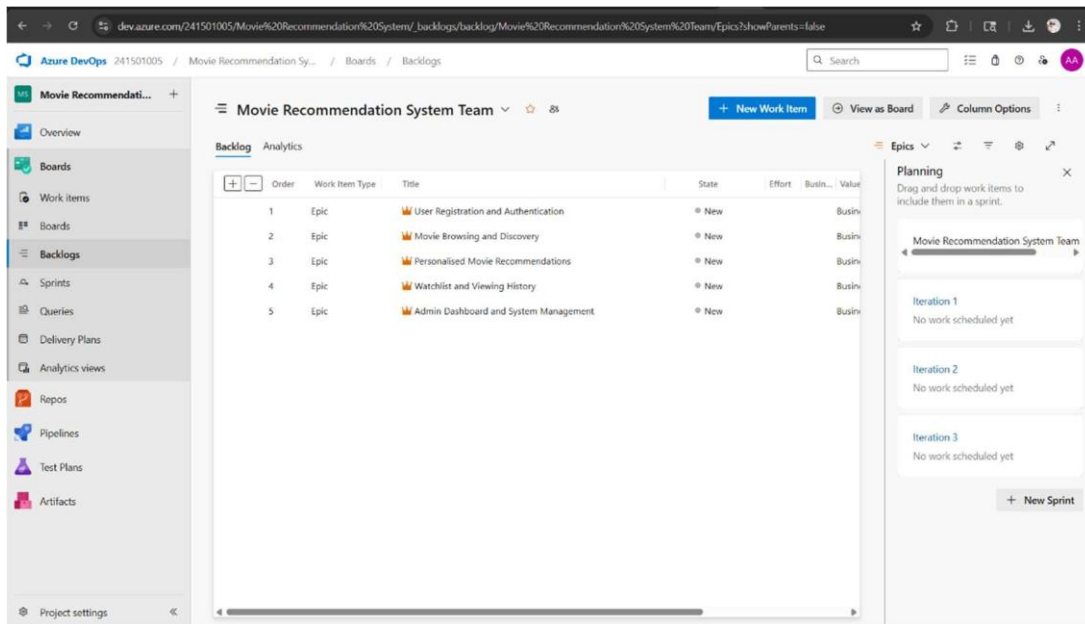
4. Project dashboard



5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or Add New Work Item depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.



## Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

Question:

In Azure platform, with a team of 4 people plan a movie recommendation system in a span of 2 sprints which includes complete user stories for all major features, including both functional (e.g., user login, movie browsing, personalised recommendations) and non-functional (e.g., reliability, scalability, performance) requirements.

Portrait the Architecture diagram / Sequence diagram / ER diagram / Class diagram in the Azure platform.

Work on the Test Cases for key functionalities and monitor test results and bug reports in Azure Test Plans. (will be updated after completion of the unit 4 and the unit 5)

## MOVIE RECOMMENDATION SYSTEM

### EPIC 1: User Registration and Authentication

#### Feature 1.1: User Registration

##### User Story 1.1a

As a user, I want to register on the Movie Recommendation System with my credentials using a strong password (minimum 8 characters, 1 numeral and 1 special character), so that I can access my personalised movie recommendations securely.

##### Task/Bug 1.1a:

- Design registration UI form.

Implement backend validation for strong password.

- Integrate password encryption (hashing).
- Create user table in Azure SQL Database.
- Send email verification link.
- Bug: Password validation not triggering for special characters.
- Bug: Duplicate email registration allowed.

##### User Story 1.1b

As a user, I want to receive an email verification link after registration so that my account is authenticated and protected from unauthorised access.

##### Task/Bug 1.1b:

- Configure email service (SMTP/Azure Communication Service).
- Generate unique verification token.
- Set token expiry (24 hours).
- Update account status after verification.
- Bug: Verification link expired immediately.
- Bug: Email sent to spam folder.

#### Feature 1.2: Login / Logout

##### User Story 1.2a

As a developer, I want to provide login to the Movie Recommendation System within 3 seconds, so that users can quickly access their personalised content.

##### Task/Bug 1.2a:

- Optimise database queries.
- Enable caching for user session data.
- Implement JWT-based session management.



- Monitor response time using Azure Monitor.
- Bug: Login takes more than 5 seconds during peak load.
- Bug: Server timeout issue.

#### User Story 1.2b

As a user, I want to log in to the Movie Recommendation System in a single step, so that I can quickly access my recommendations and watchlist.

#### Task/Bug 1.2b:

- Design simple login UI.
- Enable remember-me option.
- Implement JWT authentication.
- Provide logout functionality.
- Bug: Session not cleared after logout.
- Bug: Invalid credentials error not displayed properly.

## EPIC 2: Movie Browsing and Discovery

### Feature 2.1: Browse Movies by Genre / Language

#### User Story 2.1

As a user, I want to browse movies by genre, language, and release year so that I can discover films that match my taste.

#### Task/Bug 2.1:

- Build genre and language filter APIs.
- Store movie metadata (genre, language, year) in Azure SQL Database.
- Enable search and filter options on the movie listing page.
- Bug: Genre filter not applying correctly on page reload.

#### User Story 2.2

As a user, I want to filter movies by rating, language, and release year so that I can find movies that suit my preferences.

#### Task/Bug 2.2:

- Implement multi-select filter functionality.
- Sort movies by TMDB rating score.
- Add language and genre category tags.
- Bug: Filter state resets after page refresh.

### Feature 2.2: Movie Details Page

#### User Story 2.2.1

As a user, I want to view detailed movie information including poster, cast, synopsis and rating so that I can decide whether to watch it.

#### Task/Bug 2.2.1:

- Fetch movie poster and metadata from TMDB API.
- Display cast, director, synopsis and TMDB rating.
- Bug: Movie poster not loading for some titles.

#### User Story 2.2.2

As a user, I want to see if a movie is available on popular streaming platforms so that I know where to watch it.

#### Task/Bug 2.2.2:

- Add streaming availability field in database.

- Display streaming platform badges (e.g. Netflix, Prime).
- Bug: Incorrect streaming platform shown for some movies.

### EPIC 3: Personalised Movie Recommendations

#### Feature 3.1: Recommendation Engine

##### User Story 3.1

As a user, I want to receive movie recommendations based on my watch history and preferences so that I discover films I will enjoy.

##### Task/Bug 3.1:

- Build collaborative filtering recommendation model.
- Store user ratings and watch history in database.
- Generate top-10 personalised recommendations per user.
- Bug: Recommendations not updating after new ratings are added.

##### User Story 3.2

As a user, I want to receive mood-based movie recommendations so that I can find films suited to how I feel right now.

##### Task/Bug 3.2:

- Build mood selection UI (e.g. Happy, Thriller, Romantic).
- Map mood tags to movie genres in the backend.
- Return filtered recommendations based on selected mood.
- Bug: Mood filter returning irrelevant genres.

#### Feature 3.2: Similar Movie Suggestions

##### User Story 3.2.1

As a user, I want to see a list of similar movies on any movie's detail page so that I can find related content easily.

##### Task/Bug 3.2.1:

- Implement content-based filtering using genre and cast similarity.
- Fetch similar movie data from TMDB API.
- Display similar movies as a horizontal carousel on the detail page.
- Bug: Similar movies list showing unrelated titles.

##### User Story 3.2.2

As a user, I want to receive weekly email recommendations based on my recent activity so that I stay updated on new films.

##### Task/Bug 3.2.2:

- Build a scheduled Azure Function to send weekly recommendation emails.
- Update order status to "Paid".
- Bug: Email not triggered for users with no recent activity.

### EPIC 4: Watchlist and Viewing History

#### Feature 4.1: Watchlist Management

##### User Story 4.1

As a user, I want to add and remove movies from my watchlist so that I can keep track of films I plan to watch.

##### Task/Bug 4.1:

- Create watchlist table in Azure SQL Database.

- Build add/remove watchlist API endpoints.
- Display watchlist on user profile page.
- Bug: Watchlist not persisting after session logout.

#### **User Story 4.2**

As a user, I want to rate and review movies I have watched so that I can share my opinion and improve my recommendations.

#### **Task/Bug 4.2:**

- Build star-rating and review submission UI.
- Store ratings and reviews in the database.
- Bug: Rating submission failing silently for some users.

### **Feature 4.2: Viewing History**

#### **User Story 4.2.1**

As a user, I want to view my complete movie-watching history so that I can revisit films I have seen before.

#### **Task/Bug 4.2.1:**

- Log each movie viewed to the history table in the database.
- Display watch history with dates on user profile.
- Bug: Duplicate entries appearing in watch history.

#### **User Story 4.2.2**

As a user, I want to mark movies as “Watched” from the recommendation page so that the system can refine future suggestions.

#### **Task/Bug 4.2.2:**

- Build “Mark as Watched” button on movie cards.
- Update user history and re-trigger recommendation recalculation.
- Bug: Watched status not reflected immediately on UI.

## **EPIC 5: Admin Dashboard and System Management**

### **Feature 5.1: User Feedback and Ratings Management**

#### **User Story 5.1**

As a user, I want to provide ratings and written feedback on movies so that I can contribute to the community and help others decide.

#### **Task/Bug 5.1:**

- Build feedback form with star rating and text comment field.
- Store ratings and reviews in database linked to user and movie.
- Calculate and display average rating per movie.
- Bug: Average rating not updating after new submissions.

#### **User Story 5.2**

As an admin, I want to review and moderate user reviews so that inappropriate content is removed and service quality is maintained.

#### **Task/Bug 5.2:**

- Build admin review moderation panel in Azure DevOps.
- Flag and filter reviews below a quality threshold.
- Bug: Reviews not displaying in chronological order.

### **Feature 5.2: Movie Data and Report Management**

#### **User Story 5.2.1**

As an admin, I want to sync the latest movie data from TMDB API on a scheduled basis so that the platform always shows up-to-date content.

**Task/Bug 5.2.1:**

- CRUD operations for menu.
- Update price & availability.
- Bug: Changes not reflected immediately.

**User Story 5.2.2**

As an admin, I want to generate sales reports, so that I can analyze business performance.

**Task/Bug 5.2.2:**

- Build Azure Function to fetch new movie data daily from TMDB API.
- Export platform usage reports as PDF or Excel.
- Bug: Scheduled sync job failing silently on API timeout.

## MOVIE RECOMMENDATION SYSTEM USING MICROSOFT AZURE

*(Agile – 2 Sprints | Team of 4 | Architecture + Test Plans)*

### Agile Planning (2 Sprints)

#### Team Structure (4 Members)

| Role               | Responsibility                                      |
|--------------------|-----------------------------------------------------|
| Scrum Master       | Sprint planning, daily standups, removing blockers  |
| Backend Developer  | API, database, recommendation engine logic          |
| Frontend Developer | UI (Web/App), user experience, movie browsing pages |
| QA/Test Engineer   | Test cases, Azure Test Plans, bug tracking          |

### Sprint Planning Overview

#### Sprint 1 (Week 1–2)

**Goal:** Core user and movie browsing functionalities

Features:

- User Registration & Login
- Movie listing with genre and language filters
- Movie detail page with TMDB metadata
- Watchlist management (add/remove)

- User ratings and review submission
- Basic recommendation engine (genre-based)
- Azure SQL Database setup (Users, Movies, Ratings, Watchlist)
- Deployment on Azure App Service

## **Sprint 2 (Week 3–4)**

**Goal:** Advanced recommendations & Quality

Features:

- Personalised recommendation engine (collaborative filtering)
- Admin dashboard for data and review management
- Viewing history and watched status tracking
- Weekly email recommendations via Azure Functions
- Performance optimisation and caching
- Security enhancements (rate limiting, input validation)
- Load testing with Azure Load Testing
- Bug fixing
- CI/CD pipeline setup

## Architecture Diagram:

flowchart TD

User[User - Web / Mobile App] -->|HTTPS Request| AppService[Azure App Service - Flask API]

AppService --> Auth[Azure Active Directory - JWT Auth]

AppService --> DB[(Azure SQL Database)]

AppService --> TMDB[TMDB API - Movie Metadata]

AppService --> RecEngine[Recommendation Engine - Python]

AppService --> AzFunc[Azure Functions - Weekly Email]

AppService --> Monitor[Azure Monitor]

AppService --> DevOps[Azure DevOps CI/CD]

DevOps --> AppService

subgraph Azure Cloud

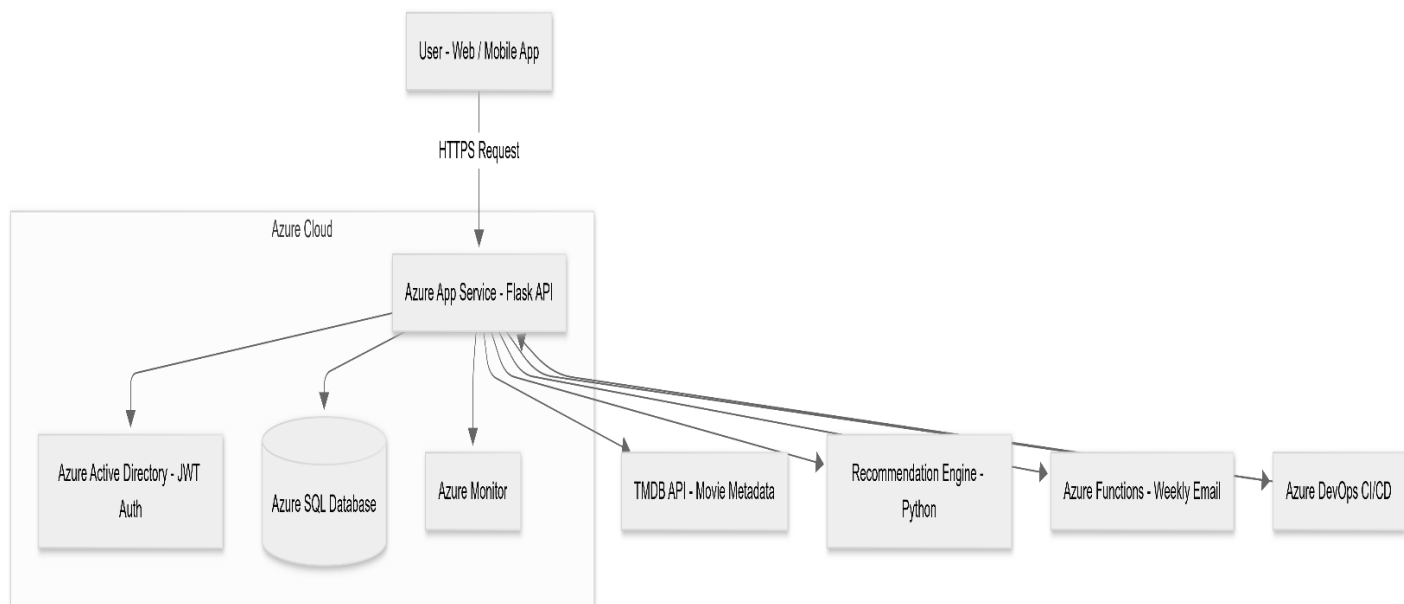
AppService

DB

Auth

Monitor

end



# Sequence Diagram

sequenceDiagram

participant User

participant UI as Web App

participant API as Flask API

participant DB as Azure SQL

participant TMDB as TMDB API

participant Rec as Recommendation Engine

User->>UI: Login

UI->>API: Authenticate Request

API->>DB: Validate Credentials

DB-->>API: User Record Found

API-->>UI: JWT Token Returned

User->>UI: Browse Movies / Apply Filters

UI->>API: GET /movies?genre=Action

API->>TMDB: Fetch Movie Metadata

TMDB-->>API: Movie List Returned

API-->>UI: Display Movie Cards

User->>UI: Request Recommendations

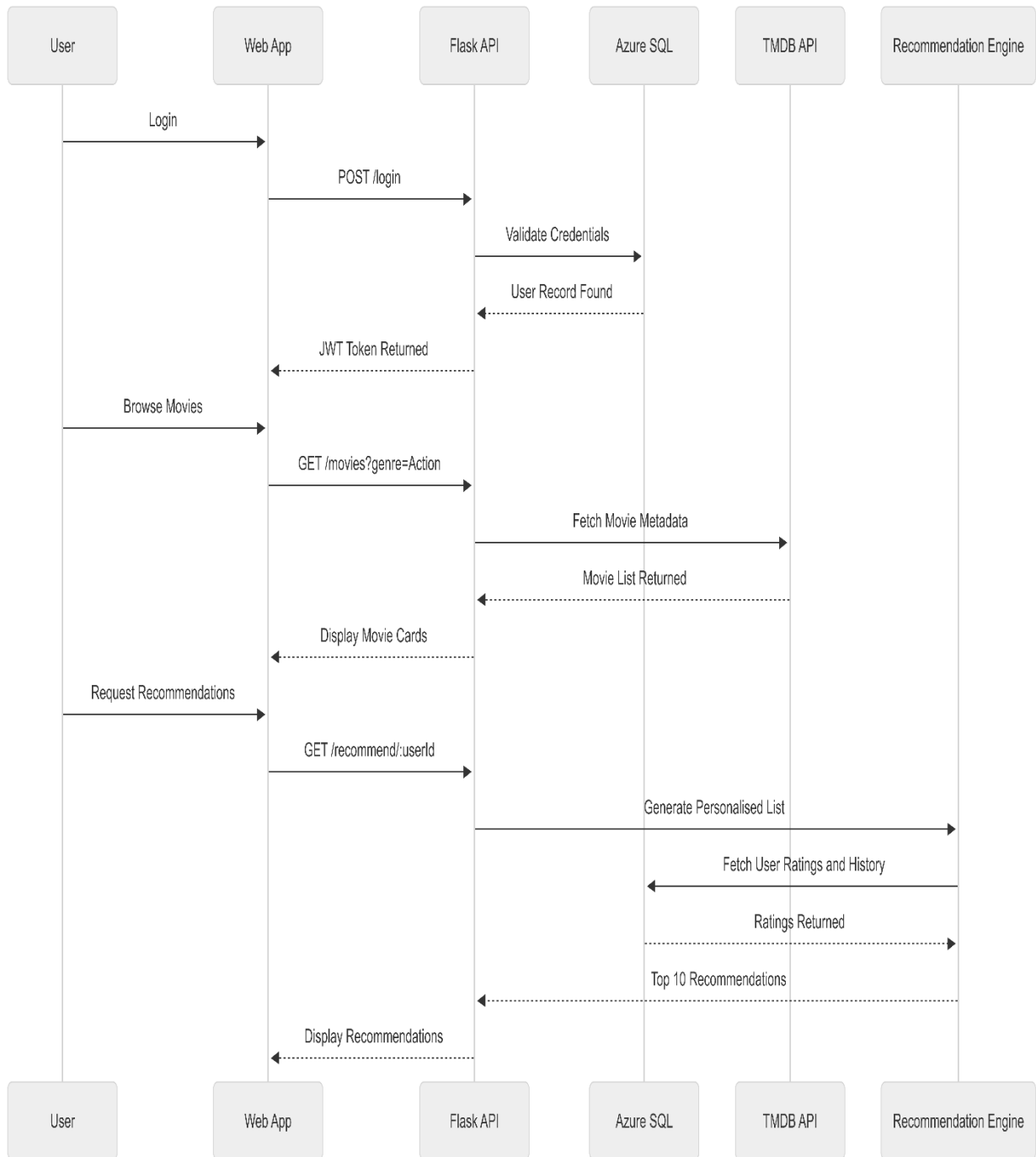
UI->>API: GET /recommend/:userId

API->>Rec: Generate Personalised List

Rec->>DB: Fetch User Ratings and History

DB-->>Rec: Ratings Returned

Rec-->>API: Top 10 Movie Recommendations





# Class Diagram

API  
classDiagram

```
class User {  
    +userId: String  
    +name: String  
    +email: String  
    +password: String  
    +watchlist: List  
    +login()  
    +logout()  
}
```

```
class Movie {  
    +movieId: String  
    +title: String  
    +genre: String  
    +rating: Float  
    +cast: List  
    +getDetails()  
    +getSimilar()  
}
```

```
class Rating {  
    +ratingId: String  
    +score: Int  
    +review: String  
    +createdAt: Date  
    +submit()  
}
```

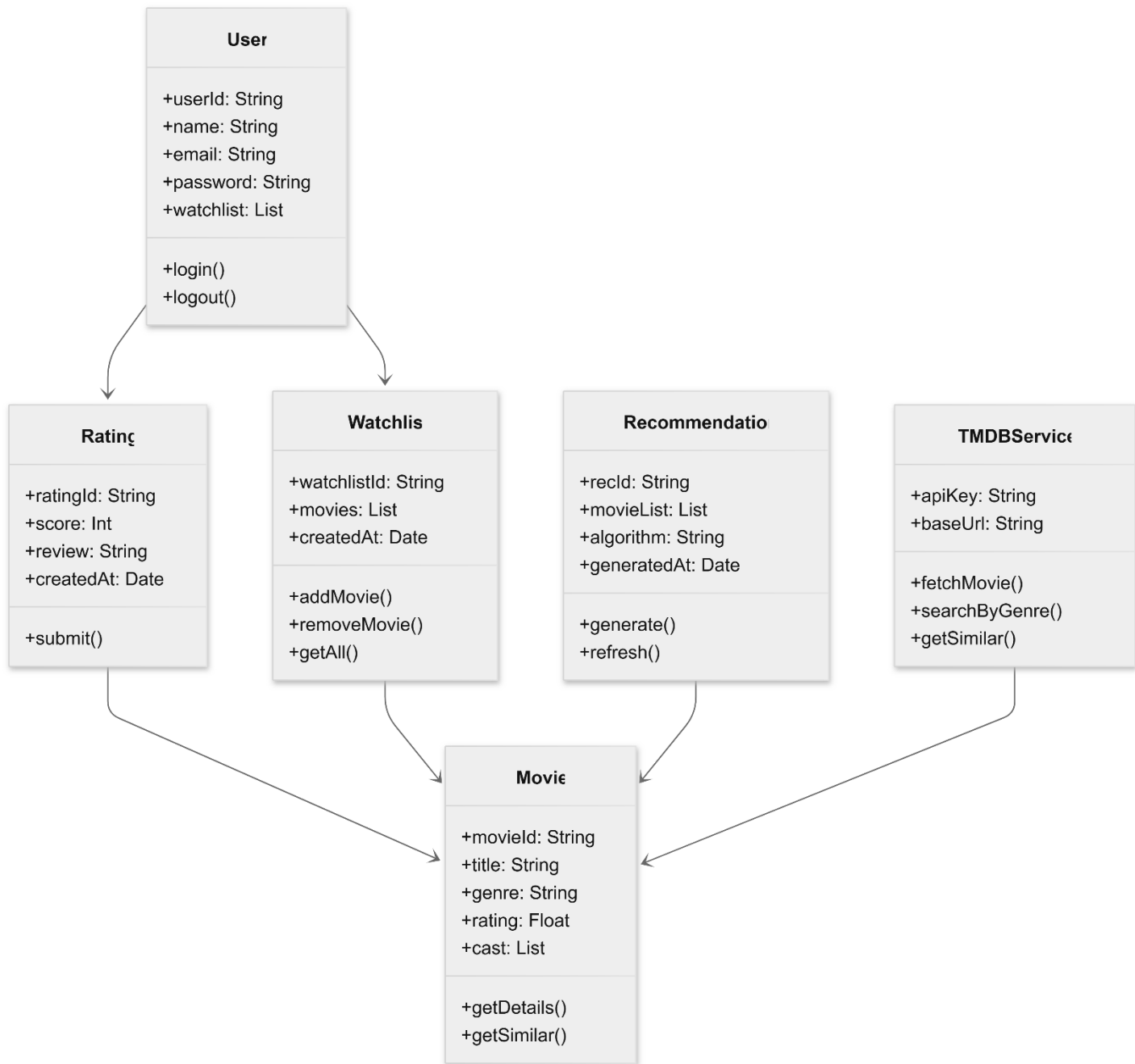
```
class Watchlist {
```

```
    +watchlistId: String  
    +movies: List  
    +createdAt: Date  
    +addMovie()  
    +removeMovie()  
    +getAll()  
}
```

```
class Recommendation {  
    +recId: String  
    +movieList: List  
    +algorithm: String  
    +generatedAt: Date  
    +generate()  
    +refresh()  
}
```

```
class TMDBService {  
    +apiKey: String  
    +baseUrl: String  
    +fetchMovie()  
    +searchByGenre()  
    +getSimilar()  
}
```

User --> Rating  
User --> Watchlist  
Rating --> Movie  
Watchlist --> Movie  
Recommendation --> Movie  
TMDBService --> Movie



## ER Diagram

A. Crow's Foot style:

- Entities are shown as boxes
- Attributes are listed inside
- Relationships use symbols like ||--o{

erDiagram

```
USER {
    int user_id PK
    string name
    string email
    string password_hash
}
```

```
MOVIE {
    int movie_id PK
    string title
    string genre
    float tmdb_rating
}
```

```
RATING {
    int rating_id PK
    int user_id FK
    int movie_id FK
    int score
    string review
}
```

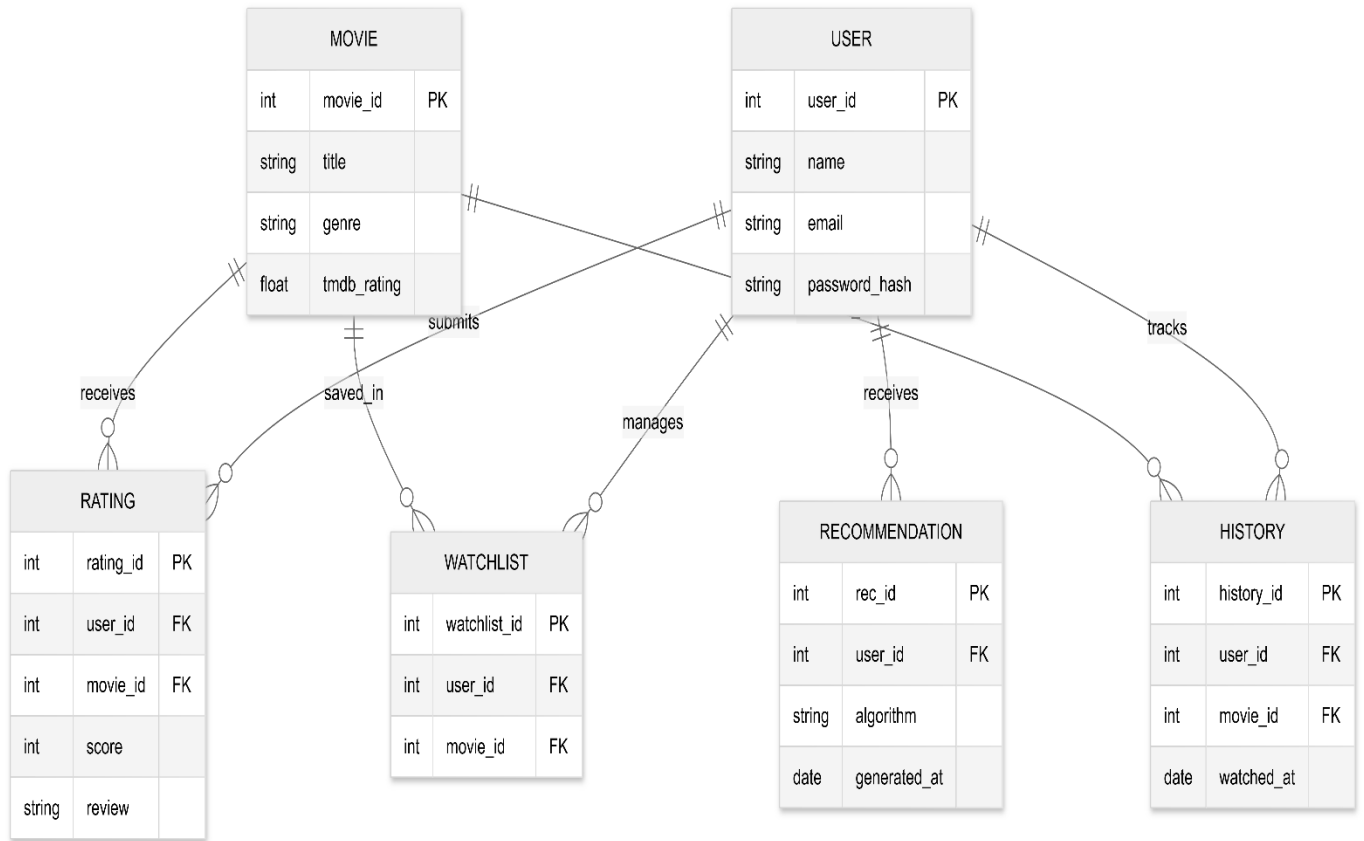
```
WATCHLIST {
    int watchlist_id PK
```

```
    int user_id FK
    int movie_id FK
}
```

```
HISTORY {
    int history_id PK
    int user_id FK
    int movie_id FK
    date watched_at
}
```

```
RECOMMENDATION {
    int rec_id PK
    int user_id FK
    string algorithm
    date generated_at
}
```

```
USER ||--o{ RATING : submits
USER ||--o{ WATCHLIST : manages
USER ||--o{ HISTORY : tracks
MOVIE ||--o{ RATING : receives
MOVIE ||--o{ WATCHLIST : saved_in
MOVIE ||--o{ HISTORY : watched_in
USER ||--o{ RECOMMENDATION : receives
```



## B. DBMS format -**Chen notation**,

-  Entity → Rectangle
-  Relationship → Diamond
-  Attributes → Ovals (circles)
-  Primary Key → Underlined
- Lines connect entity → relationship → entity
- 

flowchart LR

%% Entities

User[USER]

Movie[MOVIE]

Rating[RATING]

Watchlist[WATCHLIST]

History[HISTORY]

Recommendation[RECOMMENDATION]

%% Relationships (Diamonds)

Submits{SUBMITS}

Receives{RECEIVES}

Manages{MANAGES}

Tracks{TRACKS}

Gets{GETS}

%% Connections - mirroring the chained structure

User --> Submits

Submits --> Rating

Rating --> Receives

Receives --> Movie

User --> Manages

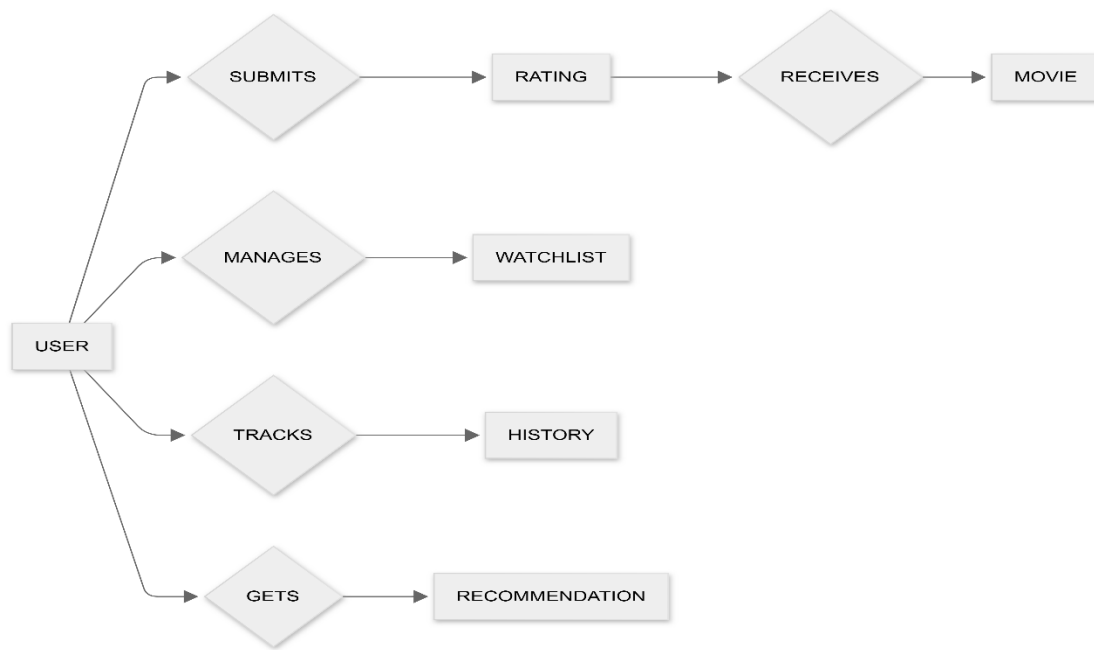
Manages --> Watchlist

User --> Tracks

Tracks --> History

User --> Gets

Gets --> Recommendation



```

SyncTMDB[Sync TMDB Movie Data]
end

```

Use case diagram:

flowchart LR

%% Actors

User((User))

Admin((Admin))

RecEngine((Recommendation Engine))

TMDBAPI((TMDB API))

AzureFunc((Azure Functions))

%% System Boundary

subgraph Movie Recommendation System

Register[Register]

Login[Login / Logout]

Browse[Browse Movies]

ViewMovie[View Movie Details]

AddWatchlist[Add to Watchlist]

RateMovie[Rate and Review Movie]

GetRecs[Get Personalised

Recommendations]

ViewHistory[View Watch History]

MoodRec[Mood-based Recommendations]

ManageMovies[Manage Movie Data]

ModerateReviews[Moderate User Reviews]

ViewReports[Generate Usage Reports]

%% User actions

User --> Register

User --> Login

User --> Browse

User --> ViewMovie

User --> AddWatchlist

User --> RateMovie

User --> GetRecs

User --> ViewHistory

User --> MoodRec

%% Admin actions

Admin --> ManageMovies

Admin --> ModerateReviews

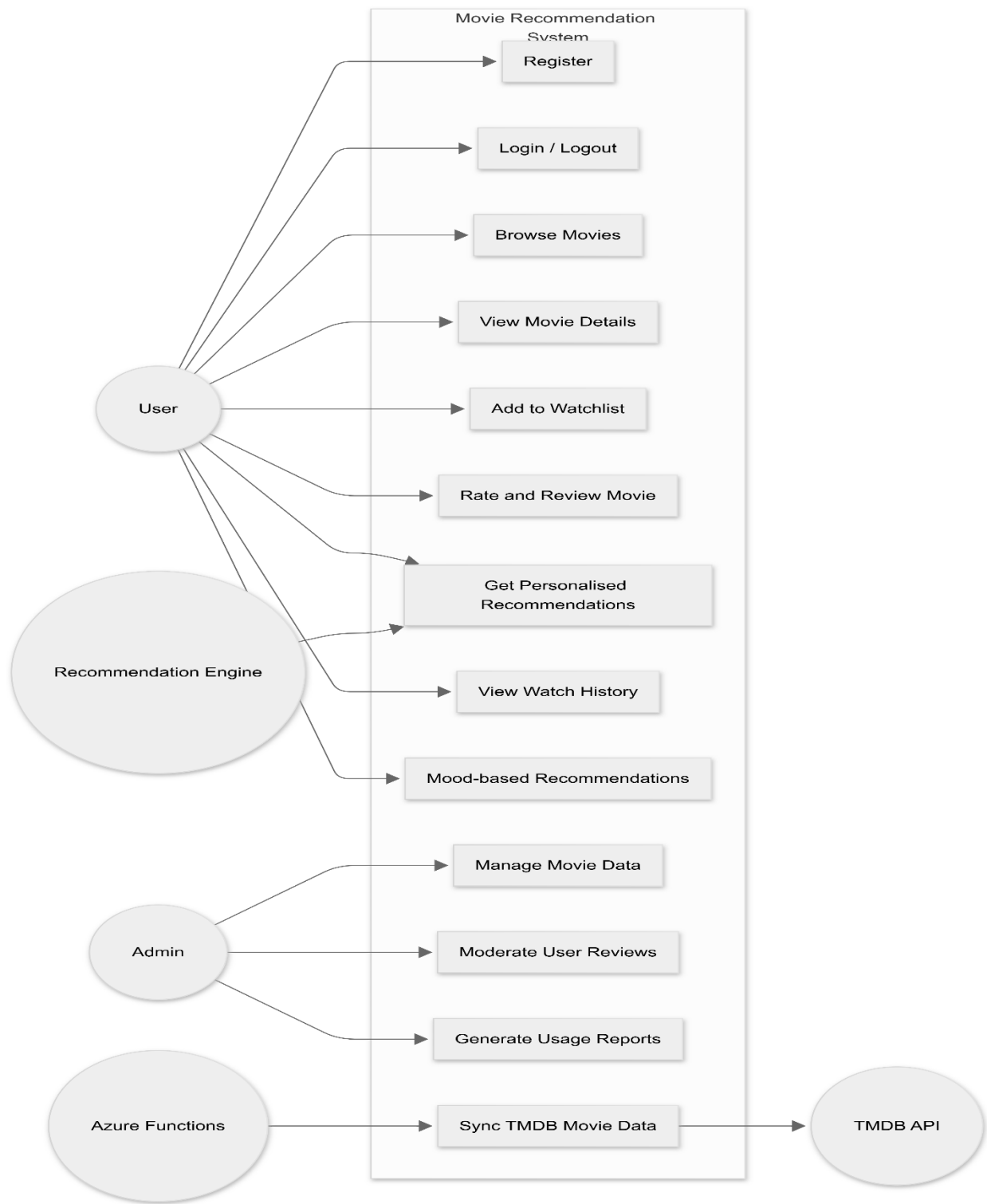
Admin --> ViewReports

%% System actions

RecEngine --> GetRecs

AzureFunc --> SyncTMDB

SyncTMDB --> TMDBAPI



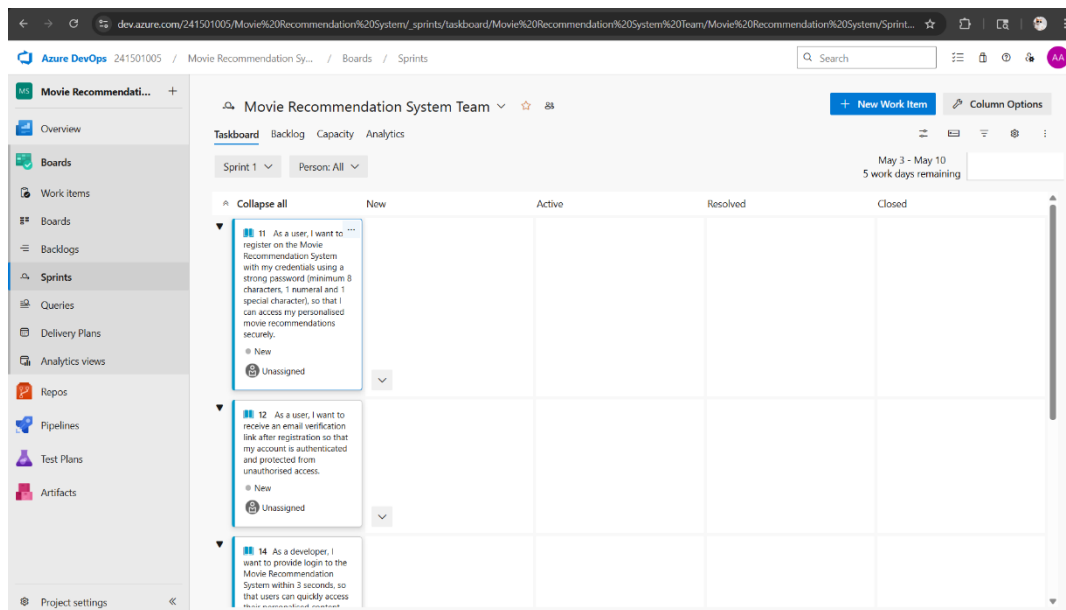
EXP NO: 4

# SPRINT PLANNING

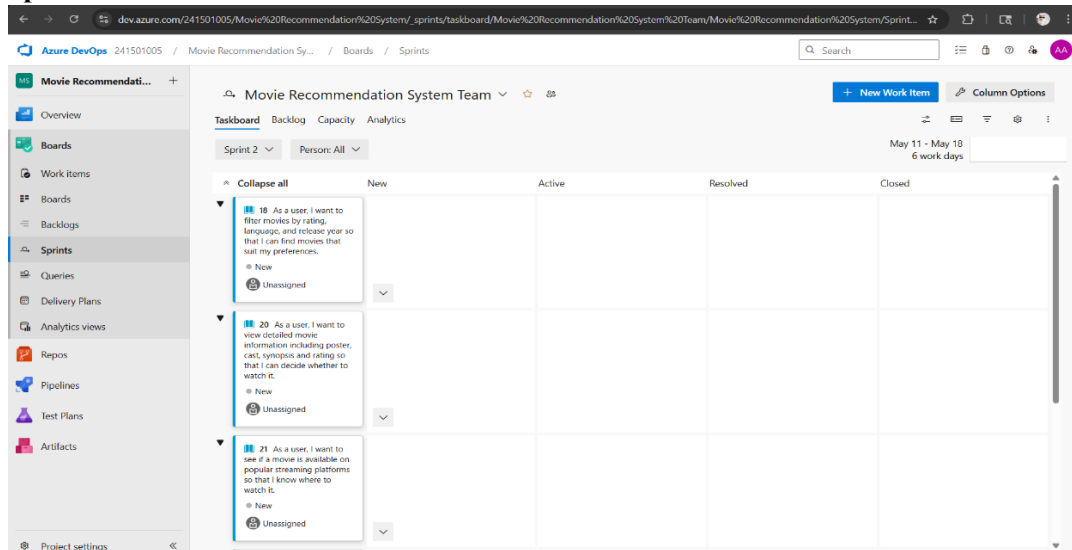
## Aim:

To assign user story to specific sprint for the Movie Recommendation System Project.

## Sprint Planning Sprint 1

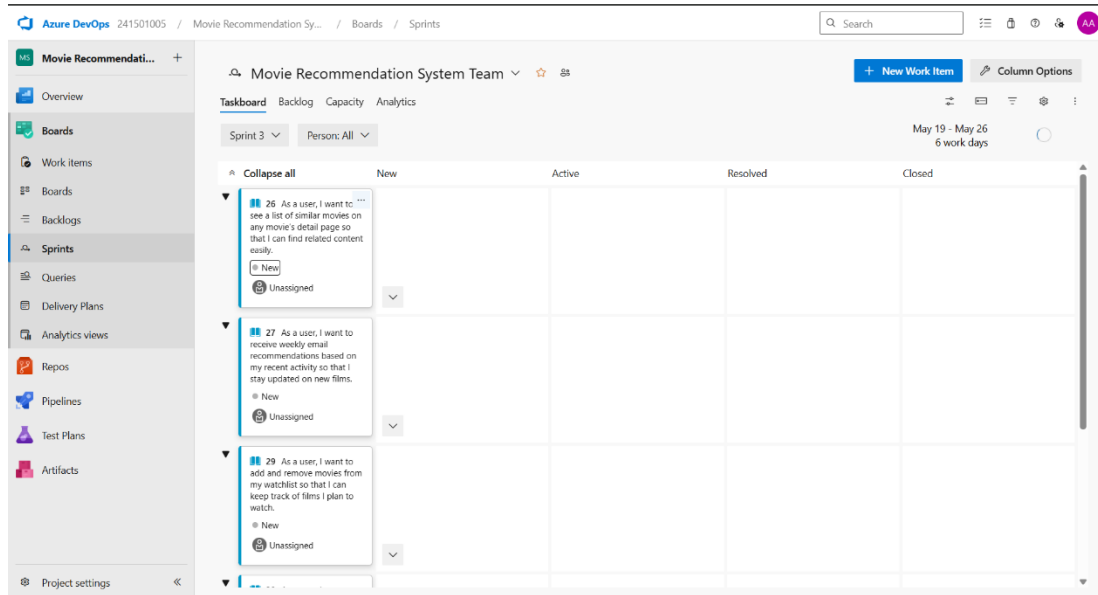


## Sprint 2





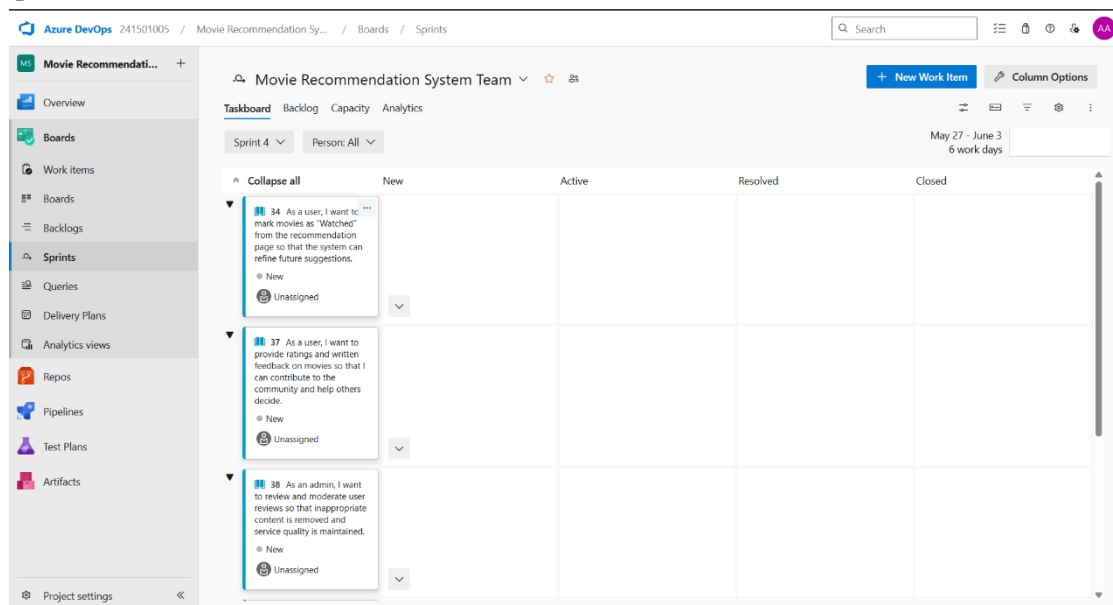
## Sprint 3



The screenshot shows the Azure DevOps interface for a project named "Movie Recommendation System". The left sidebar contains navigation links: Overview, Boards, Work Items, Backlogs, Sprints (selected), Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main area displays the "Sprints" board for "Sprint 3", which runs from May 19 to May 26 (6 work days). The board is organized into columns: New, Active, Resolved, and Closed. Three work items are currently in the "New" column:

- Item 26: "As a user, I want to see a list of similar movies on any movie's detail page so that I can find related content easily." (New, Unassigned)
- Item 27: "As a user, I want to receive weekly email recommendations based on my recent activity so that I stay updated on new films." (New, Unassigned)
- Item 29: "As a user, I want to add and remove movies from my watchlist so that I can keep track of films I plan to watch." (New, Unassigned)

## Sprint 4



The screenshot shows the Azure DevOps interface for the same project, "Movie Recommendation System". The left sidebar is identical to the previous screenshot. The main area displays the "Sprints" board for "Sprint 4", which runs from May 27 to June 3 (6 work days). The board is organized into columns: New, Active, Resolved, and Closed. Three work items are currently in the "New" column:

- Item 34: "As a user, I want to mark movies as 'Watched' from the recommendation page so that the system can refine future suggestions." (New, Unassigned)
- Item 37: "As a user, I want to provide ratings and written feedback on movies so that I can contribute to the community and help others decide." (New, Unassigned)
- Item 38: "As an admin, I want to review and moderate user reviews so that inappropriate content is removed and service quality is maintained." (New, Unassigned)

## Result:

The Sprints are created for the Movie Recommendation System Project.

EXP NO: 5

# POKER ESTIMATION

## Aim:

Create Poker Estimation for the user stories – Movie Recommendation System Project.

## Poker Estimation

USER STORY 17

17 As a user, I want to browse movies by genre, language, and release year so that I can discover films that match my taste.

No one selected 0 Comments Add Tag

Save and Close Follow

Updated by Abdulla Sabith A: Just now

State: New Area: Movie Recommendation System Reason: New Iteration: Movie Recommendation System\Sprint 1

Details 1 0

**Description**

Click to add Description.

**Acceptance Criteria**

Click to add Acceptance Criteria.

**Discussion**

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

**Planning**

Story Points  
5  
Priority  
2  
Risk

**Classification**

Value area  
Business

**Deployment**

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

**Development**

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

**Related Work**

Add link

Parent

16 Browse Movies by Genre / Language  
Updated 1h ago New

## Result:

The Estimation/Story Points is created for the project using Poker Estimation.

**EXP NO: 8**

## **TESTING – TEST PLANS AND TEST CASES**

### **Aim:**

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

### **1.New test plan**

Azure DevOps 241501005 / Movie Recommendation Sy... / Test Plans

Search

Movie Recommendation System - Test Plan

Area Path \*  
Movie Recommendation System

Iteration \*  
Movie Recommendation System

Create Cancel

### **2.Test suite**

Azure DevOps 241501005 / Movie Recommendation Sy... / Test Plans / Movie Recommendation Sy...

Search

Movie Recommendation System - Test Plan

Test Suites

Filter suites by name

Movie Recommendation System - Test Plan

- New Suite
- Watchlist Management
- Recommendation System
- Movie Details
- Movie Browsing & Filtering
- User Registration & Login (1)

User Registration & Login (ID: 43)

Define Execute Chart

Test Cases (1 item)

| Title                   | Order | Test Case Id | Assigned To          | State |
|-------------------------|-------|--------------|----------------------|-------|
| Successful Registration | 1     | 48           | Abdulrahman A. Desig | Pass  |

### 3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Movie Recommendation System – Test Plans

#### USER STORIES

- As a user, I want to sign up and log in securely so that I can access my movie recommendations.
- As a user, I want to log in to the Movie Recommendation System in a single step.
- As a user, I want to browse movies by genre, language, and release year.
- As a user, I want to receive movie recommendations based on my watch history and preferences.
- As a user, I want to add and remove movies from my watchlist.

#### Test Suites

##### Test Suit: TS01 - User Registration (ID: 86)

###### 1. TC01 – Successful Registration

o Action:

- Go to the Registration page.
- Enter valid username, email, and password.
- Click "Register".

o Expected Results:

- Registration page is displayed.
- Fields accept values without error.
- Account is created successfully and email verification is sent.

o Type: Happy Path

###### 2. TC02 – Weak Password

o Action:

- Go to the Registration page.
- Enter password without special character.
- Click "Register".

o Expected Results:

- Fields accept values without error.
- Error message "Password does not meet criteria" is displayed.

o Type: Error Path

---

##### Test Suit: TS02 - Login (ID: 87)

1. **TC03 – Successful Login**

o Action:

- Go to the Login page.
- Enter valid email and password.
- Click "Login".

o Expected Results:

- Login form is displayed.
- Fields accept data without error.
- User is logged in and redirected to the dashboard.

o Type: Happy Path

2. **TC04 – Invalid Credentials**

o Action:

- Go to the Login page.
- Enter wrong password.
- Click "Login".

o Expected Results:

- Fields accept data.
- Error message "Invalid credentials" is displayed.

o Type: Error Path

---

**Test Suit: TS03 - Movie Browsing (ID: 88)**

1. **TC05 – Browse Movies**

o Action:

- Log in successfully.
- Navigate to Movies section.

o Expected Results:

- Movies are displayed clearly.

o Type: Happy Path

2. **TC06 – Filter Failure**

o Action:

- Apply invalid filter.
- Refresh results.

- o Expected Results:
  - No movies are displayed.
  - Message "No movies found" is shown.

- o Type: Error Path

---

#### **Test Suit: TS04 - Recommendation System (ID: 89)**

##### **1. TC07 – Personalized Recommendations**

- o Action:
  - Login with valid credentials.
  - Open Recommendations section.
- o Expected Results:
  - Movies are recommended based on user history.

- o Type: Happy Path

##### **2. TC08 – No History Case**

- o Action:
  - Login as new user.
  - Open Recommendations section.
- o Expected Results:
  - Default recommendations are displayed.

- o Type: Error Path

---

#### **Test Suit: TS05 - Watchlist Management (ID: 90)**

##### **1. TC09 – Add to Watchlist**

- o Action:
  - Login successfully.
  - Select a movie.
  - Click "Add to Watchlist".
- o Expected Results:
  - Movie is added successfully to watchlist.

- o Type: Happy Path

## 2. TC10 – Duplicate Add

### o Action:

- Select a movie already added.
- Click "Add to Watchlist" again.

### o Expected Results:

- Duplicate addition is prevented.
- Message indicating duplicate is shown.

### o Type: Error Path

## Test Cases

The screenshot displays the Azure DevOps Test Plans interface. The left sidebar shows the navigation menu with 'Test Plans' selected. The main area shows a test suite named 'User Registration & Login (1)' with a 'New Test Case' button. Below the button, a table lists the test cases:

| Title                   | Order | Test Case Id | Assigned To            | State |
|-------------------------|-------|--------------|------------------------|-------|
| Successful Registration | 1     | 40           | Abdulla Subith A Desig |       |

dev.azure.com/241501005/Movie%20Recommendation%20System/\_testPlans/define?planId=418&suiteId=43

TEST CASE 50

50 Weak Password

Abdulla Sabith A 0 Comments Add Tag

Save and Close Follow

Updated by Abdulla Sabith A: Just now

State: Design Area: Movie Recommendation System

Reason: New Iteration: Movie Recommendation System

Steps Summary Associated Automation

Steps

| Step | Action                                   | Expected result                        | Attachments |
|------|------------------------------------------|----------------------------------------|-------------|
| 1.   | Enter password without special character | Error: password does not meet criteria |             |
|      | Click register                           | Type: Error Path                       |             |
| 2.   | Click or type here to add a step.        |                                        |             |

Parameter values

Status

Priority: 2

Automation status: Not Automated

Custom

Type:

## 4.Installation of test

chromewebstore.google.com/detail/test-feedback/gnldpbncfnlkkicnapihmaphldnldjlb?utm\_source=ext\_app\_menu

chrome web store Search extensions and themes

Discover Extensions Themes

 **Test & Feedback** Add to Chrome

4.2 ★ (176 ratings) Share

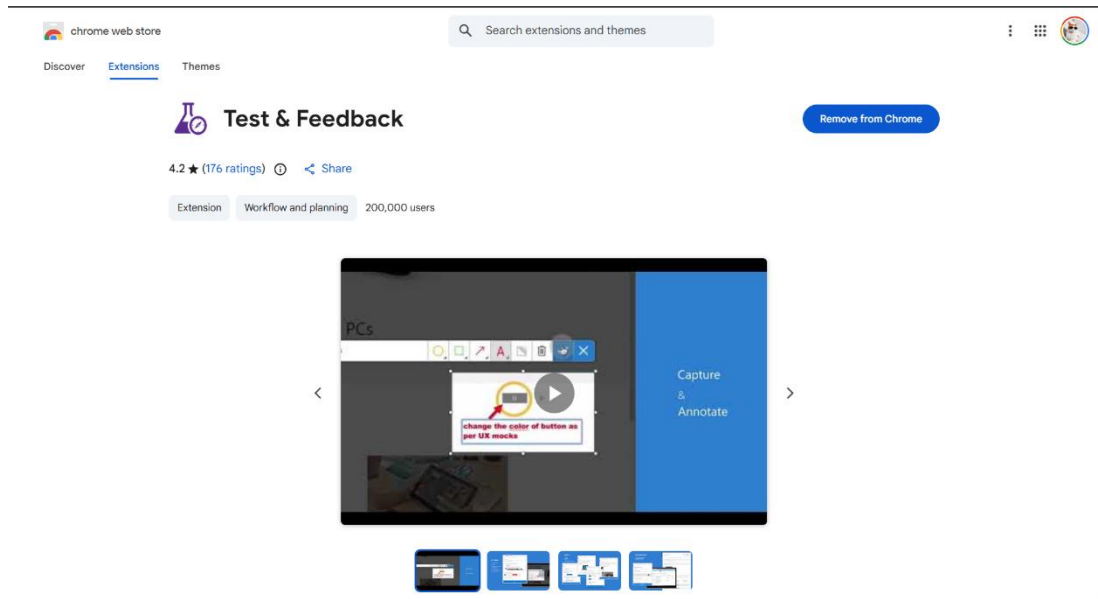
Extension Workflow and planning 200,000 users

PCs

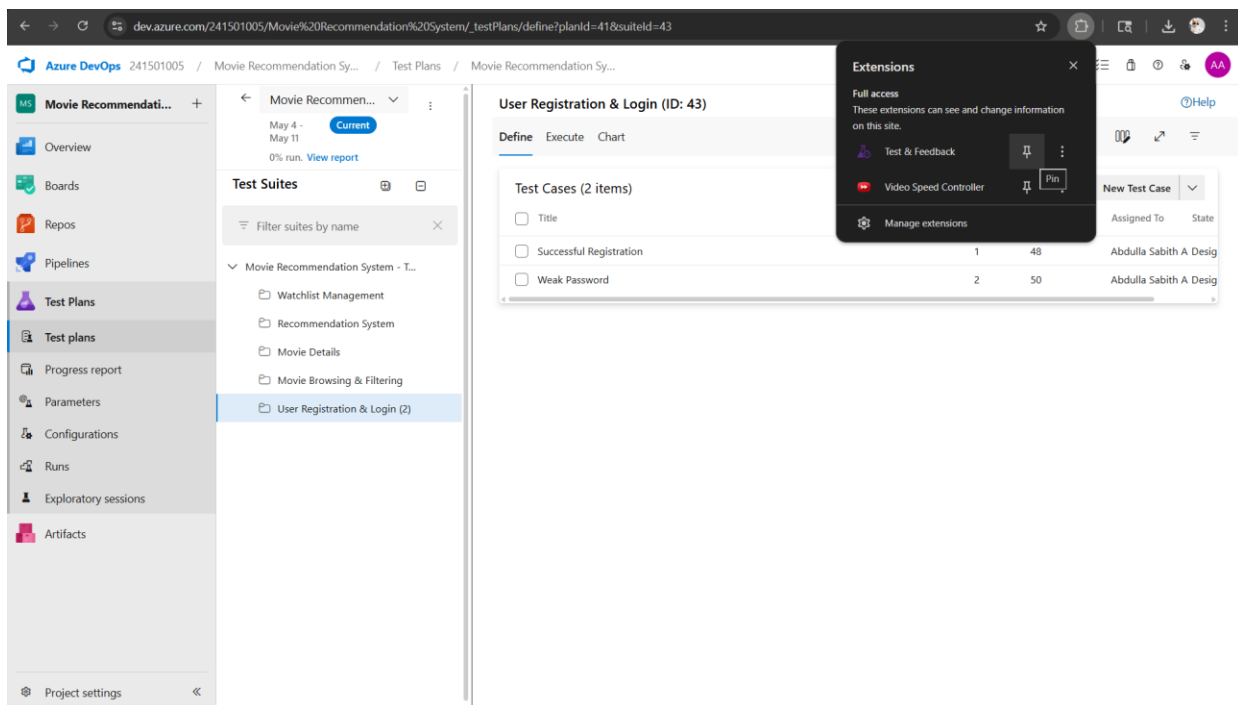
Capture & Annotate

change the color of button as per UX mock

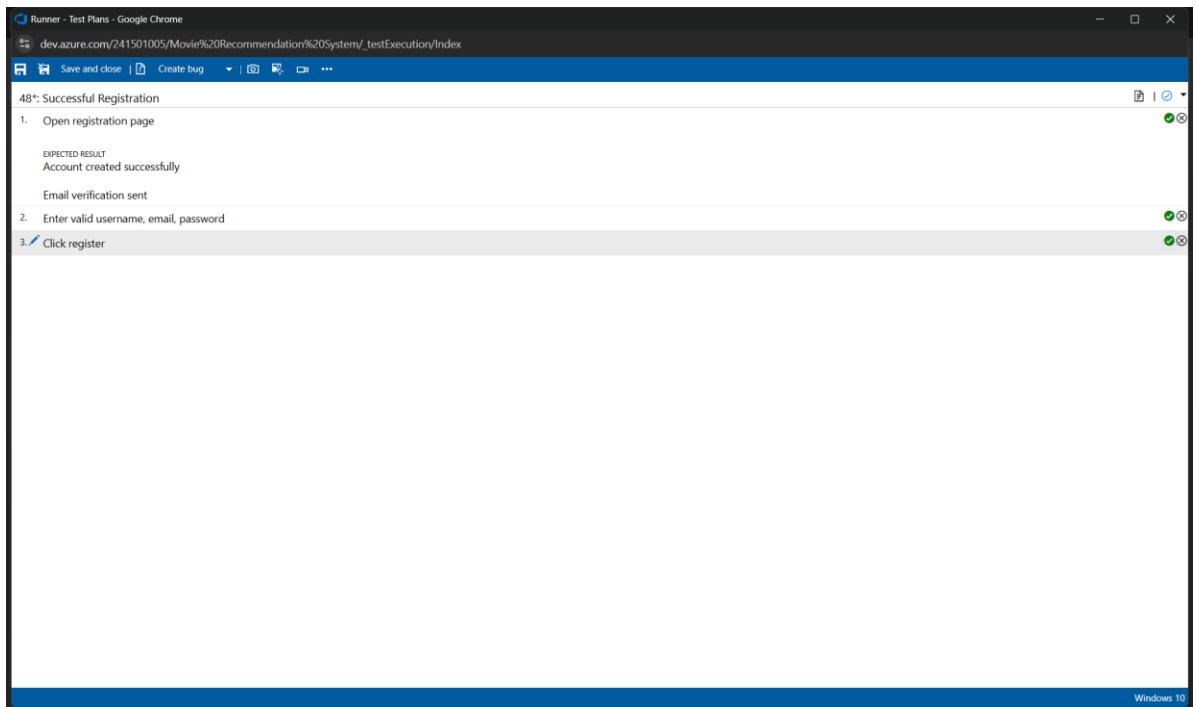
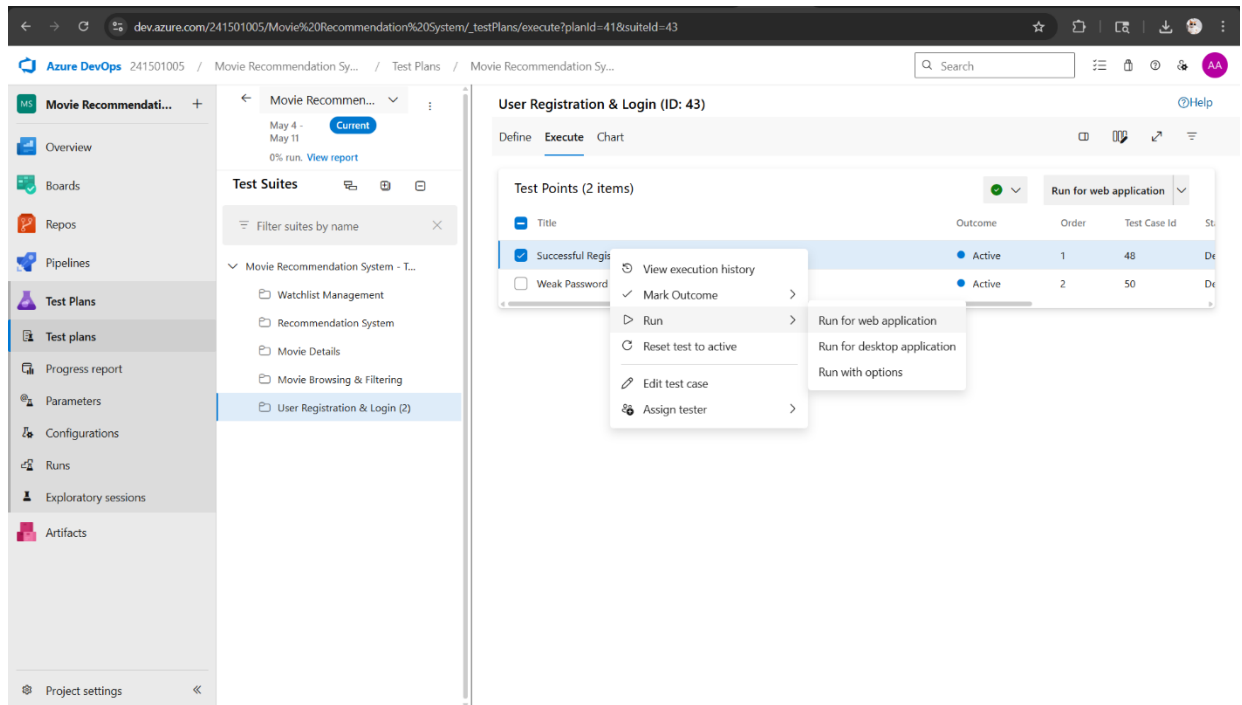




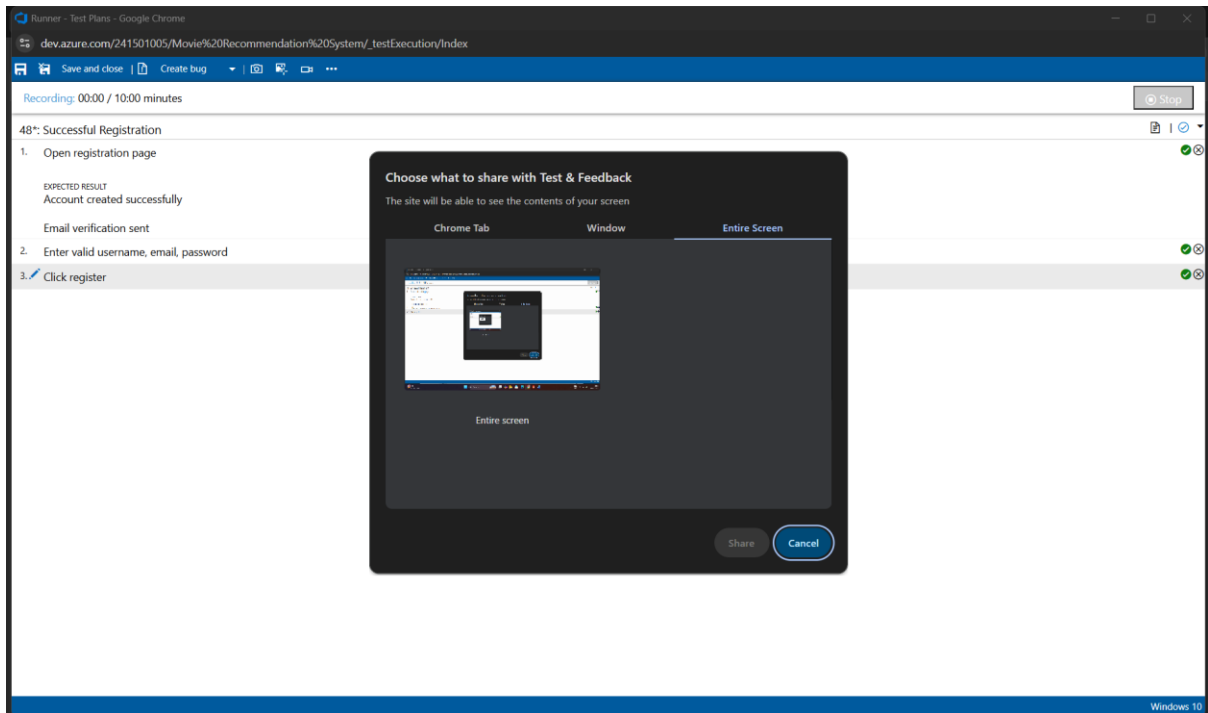
Test and feedback  
Showing it as an extension



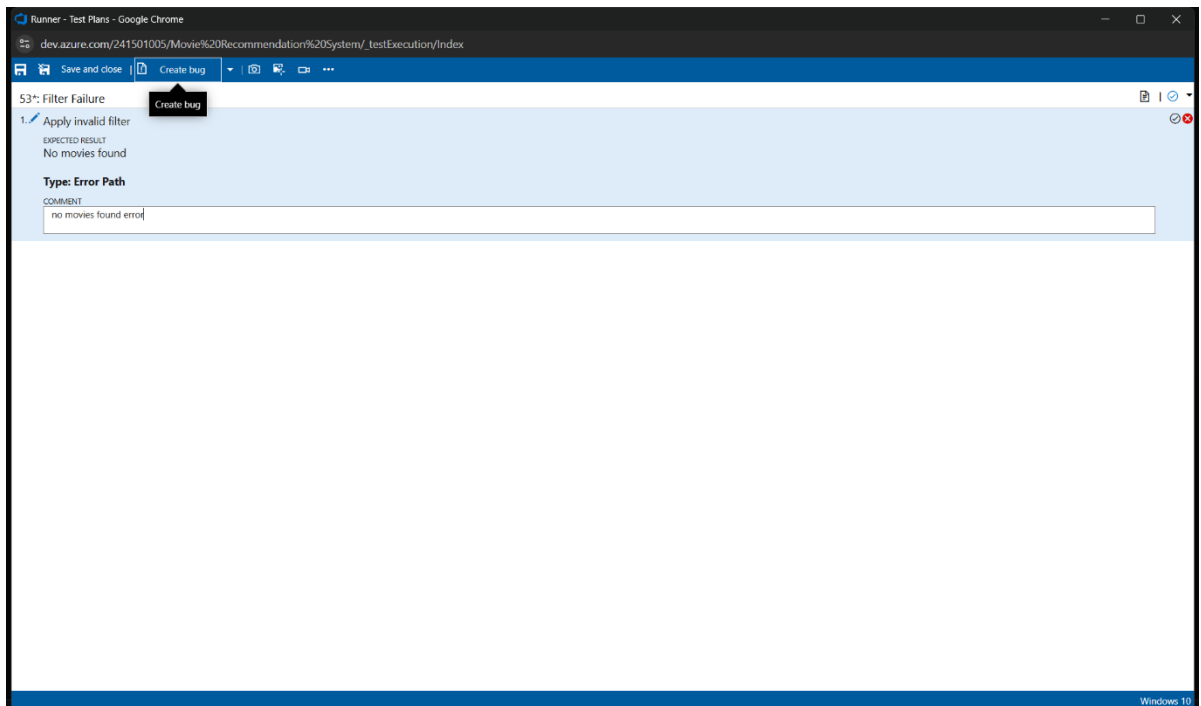
## 5. Running the test cases



## 6. Recording the test case



## 7. Creating the bug



Runner - Test Plans - Google Chrome

dev.azure.com/241501005/Movie%20Recommendation%20System/\_testExecution/Index

NEW BUG \*

Movies not found error

No one selected 0 Comments Add Tag

Save and Close

State: New Area: Movie Recommendation System Reason: New Iteration: Movie Recommendation System

Details 2 0

**Repro Steps**

5/4/2026 12:02 AM Bug filed on "Filter Failure"

| Step no. | Result | Title                |
|----------|--------|----------------------|
| 1.       | Failed | Apply invalid filter |

Expected Result

No movies found

Type: Error Path

Comment

no movies found error

Test Configuration: Windows 10

**System Info**

|                                 |                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Browser - Name                  | Google Chrome 147                                                                                               |
| Browser - Language              | en-US                                                                                                           |
| Browser - Height                | 912                                                                                                             |
| Browser - Width                 | 1536                                                                                                            |
| Browser - User agent            | Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 |
| Operating system - Name         | Windows NT 10.0; Win64; x64                                                                                     |
| Operating system - Architecture | x86_64                                                                                                          |
| Operating system - Processor    | 12th Gen Intel(R) Core(TM) i5-12500H                                                                            |

**Planning**

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

**Effort (Hours)**

Original Estimate

Remaining

Completed

**Deployment**

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

**Development**

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also create a branch to get started.

**Related Work**

Add link

Add an existing work item as a parent

Tested By

53 Filter Failure

Updated Just now @ Design

**System Info**

Found in Build

Runner - Test Plans - Google Chrome

dev.azure.com/241501005/Movie%20Recommendation%20System/\_testExecution/Index

NEW BUG \*

Movies not found error

No one selected 0 Comments Add Tag

Save and Close

State: New Area: Movie Recommendation System Reason: New Iteration: Movie Recommendation System

Details 2 0

**System Info**

|                                         |                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Browser - Name                          | Google Chrome 147                                                                                               |
| Browser - Language                      | en-US                                                                                                           |
| Browser - Height                        | 912                                                                                                             |
| Browser - Width                         | 1536                                                                                                            |
| Browser - User agent                    | Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 |
| Operating system - Name                 | Windows NT 10.0; Win64; x64                                                                                     |
| Operating system - Architecture         | x86_64                                                                                                          |
| Operating system - Processor            | 12th Gen Intel(R) Core(TM) i5-12500H                                                                            |
| Operating system - Number of processors | 16                                                                                                              |
| Memory - Available                      | 4847112192                                                                                                      |
| Memory - Capacity                       | 16833748992                                                                                                     |
| Display - Pixels per inch (X axis)      | 120                                                                                                             |
| Display - Pixels per inch (Y axis)      | 120                                                                                                             |
| Display - Device pixel ratio            | 1.25                                                                                                            |

Completed

Tested By

53 Filter Failure

Updated Just now @ Design

**System Info**

Found in Build

Integrated in Build

**Discussion**

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[Switch to Markdown editor](#)

## 8. Test case results

The screenshot shows the Azure DevOps interface for a failed test run. The left sidebar contains navigation options: Overview, Boards, Repos, Pipelines, Test Plans, Runs, and Artifacts. The main content area displays the 'Run - 4 Movie Browsin...' with a '53 Filter Failure' status. The 'Summary' section includes details about the run by Abdulla Sabith A, pipeline run tested (None), configuration (Windows 10), completed time (5/4/2026 12:03 AM), test plan (41 Movie Recommendation System - Test), test suite (44 Movie Browsing & Filtering), test case (53 Filter Failure), and priority (2). The 'Analysis' section shows the analysis owner as Abdulla Sabith A, with failure type, resolution, and comment all set to 'None'. Below this, the 'Linked work items' section shows a table with one item: ID 56, Work Item title 'Movies not found error', State 'New', Assigned To 'Unassigned', Created date '5m ago', and Updated date '5m ago'. The 'Test failed' section shows the start time (5/4/2026 12:00 AM), duration (2m 34s), and an attachment 'SystemInformation-2026-05-04T00-02-05.246Z.json (534 B)'. The 'Attachments' section shows a table with one row: Step 1, Outcome 'Failed', Action 'Apply invalid filter', and Expected Result 'No movies found'.

## 9. Test report summary

The screenshot shows the Azure DevOps interface for a work item. The left sidebar contains navigation options: Overview, Boards, Work Items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main content area displays the 'Work Items' section with a '56 Movies not found error' status. The 'Recently created' section shows the work item with a 'BUG 56' label. The 'Details' section includes the state (New), area (Movie Recommendation System), reason (New), and iteration (Movie Recommendation System). The 'Repro Steps' section shows the steps to reproduce the issue: 1. Apply invalid filter. The 'Expected Result' is 'No movies found' and the 'Type: Error Path'. The 'Test Configuration' section shows the configuration (Windows 10). The 'System Info' section shows the browser details: Browser - Name (Google Chrome 147), Browser - Language (en-US), and Browser - Height (912). The 'Planning' section shows the resolved reason, story points, priority (2), severity (3 - Medium), and activity. The 'Deployment' section shows the deployment status and a link to the deployment status reporting. The 'Development' section shows the development status and a link to the development status reporting. The 'Related Work' section shows a link to the related work item.

- Assigning bug to the developer and changing state

dev.azure.com/241501005/Movie%20Recommendation%20System/\_workitems/edit/56/

Azure DevOps 241501005 / Movie Recommendation Sy... / Boards / Work items

Movie Recommendation... +

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Recently created

Back to Work Items

1 of 49

BUG 56

56 Movies not found error

No one selected

0 Comments Add Tag

Save Follow

Updated by Abdulla Sabith A: 7m ago

Details

State: New

Area: Movie Recommendation System

Reason: New

Iteration: Movie Recommendation System

System Info

|                                         |                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Browser - Name                          | Google Chrome 147                                                                                               |
| Browser - Language                      | en-US                                                                                                           |
| Browser - Height                        | 912                                                                                                             |
| Browser - Width                         | 1536                                                                                                            |
| Browser - User agent                    | Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 |
| Operating system - Name                 | Windows NT 10.0; Win64; x64                                                                                     |
| Operating system - Architecture         | x86_64                                                                                                          |
| Operating system - Processor model      | 12th Gen Intel(R) Core(TM) i5-12500H                                                                            |
| Operating system - Number of processors | 16                                                                                                              |
| Memory - Available                      | 4847112192                                                                                                      |
| Memory - Capacity                       | 16833748992                                                                                                     |
| Display - Pixels per inch (X axis)      | 120                                                                                                             |
| Display - Pixels per inch (Y axis)      | 120                                                                                                             |
| Display - Device pixel ratio            | 1.25                                                                                                            |

Remaining

Completed

Related Work

Add link

Add an existing work item as a parent

Tested By

53 Filter Failure

Updated 10m ago @ Design

System Info

Found in Build

Integrated in Build

Discussion

## 10. Progress report

dev.azure.com/241501005/Movie%20Recommendation%20System/\_testManagement/analytics/progressreport

Azure DevOps 241501005 / Movie Recommendation Sy... / Test Plans / Progress report

Movie Recommendation... +

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Parameters

Configurations

Runs

Exploratory sessions

Artifacts

Project settings

Progress report

Movie Recommendation System - Test Plan

Test Suites

Outcome

Configuration

Tester

Priority

Assigned To

Summary

1 Test plans

6 Test points

6 (6 / 6) Test points run

100% Run

83% (5 / 6) Pass rate

5 Passed

1 Failed

Outcome trend

5/1/2026 - 5/30/2026

Tests

2026-05-01

2026-05-02

2026-05-03

2026-05-04

Not run

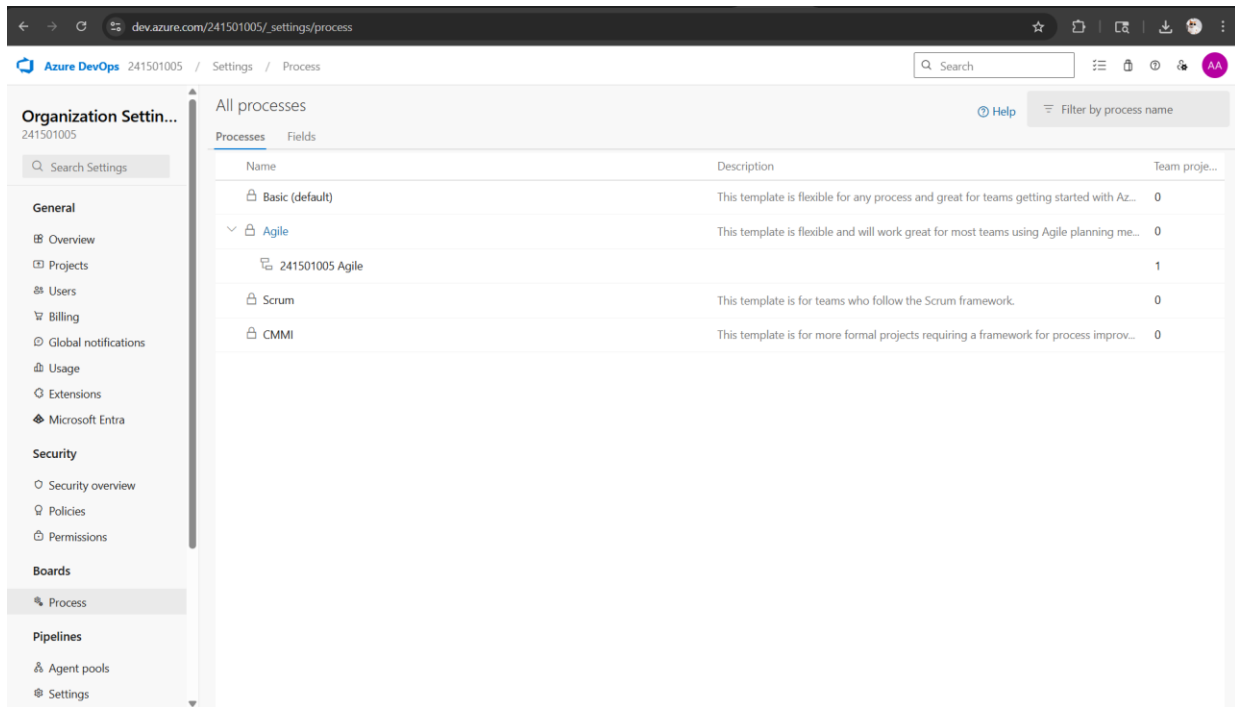
Passed

Failed

Details

| Test plan name | Test points | Run % | Passed % | Failed % | Not run count |
|----------------|-------------|-------|----------|----------|---------------|
|                |             |       |          |          |               |

## 11. Changing the test template



### Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9

## LOAD TESTING AND PERFORMANCE TESTING

### Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint.

### Load Testing

#### Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal  
Go to <https://portal.azure.com> and log in.
2. Create the Resource ○ Go to *Create a resource* → Search for “Azure Load Testing”.
  - Select Azure Load Testing and click Create.
3. Fill in the Configuration Details ○
  - Subscription: Choose your Azure subscription. ○
  - Resource Group: Create new or select an existing one. ○
  - Name: Provide a unique name (no special characters). ○
  - Location: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

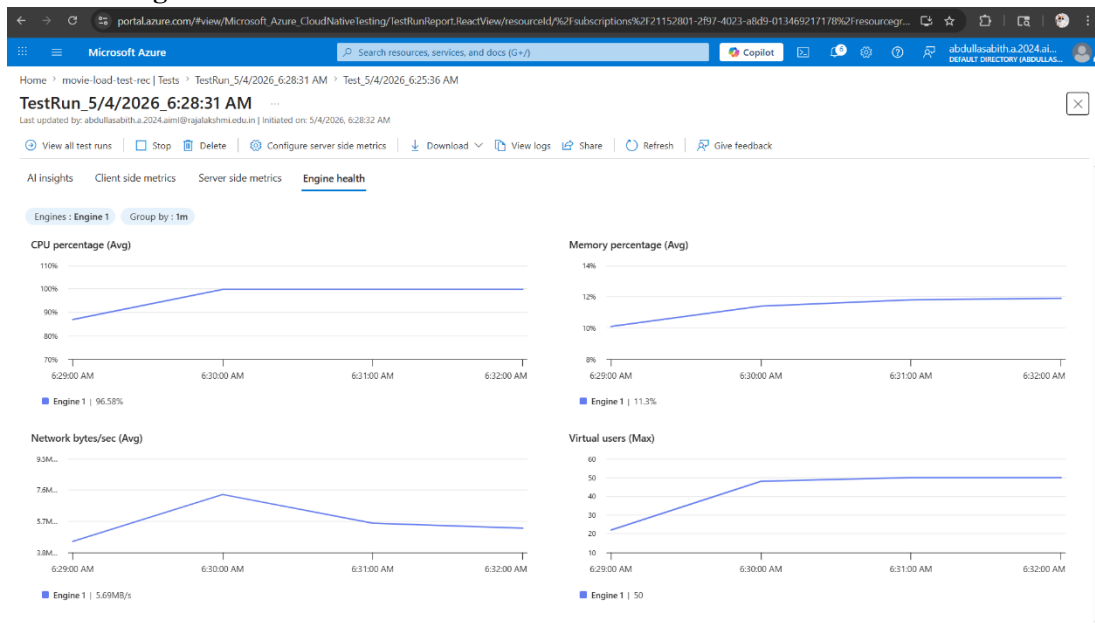
#### Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab ○
  - Test Name: Provide a unique name.
  - Description: (Optional) Add test purpose.
  - Run After Creation: Keep checked.
3. Load Settings ○
  - Test URL: Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.



## Load Testing



### Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint.

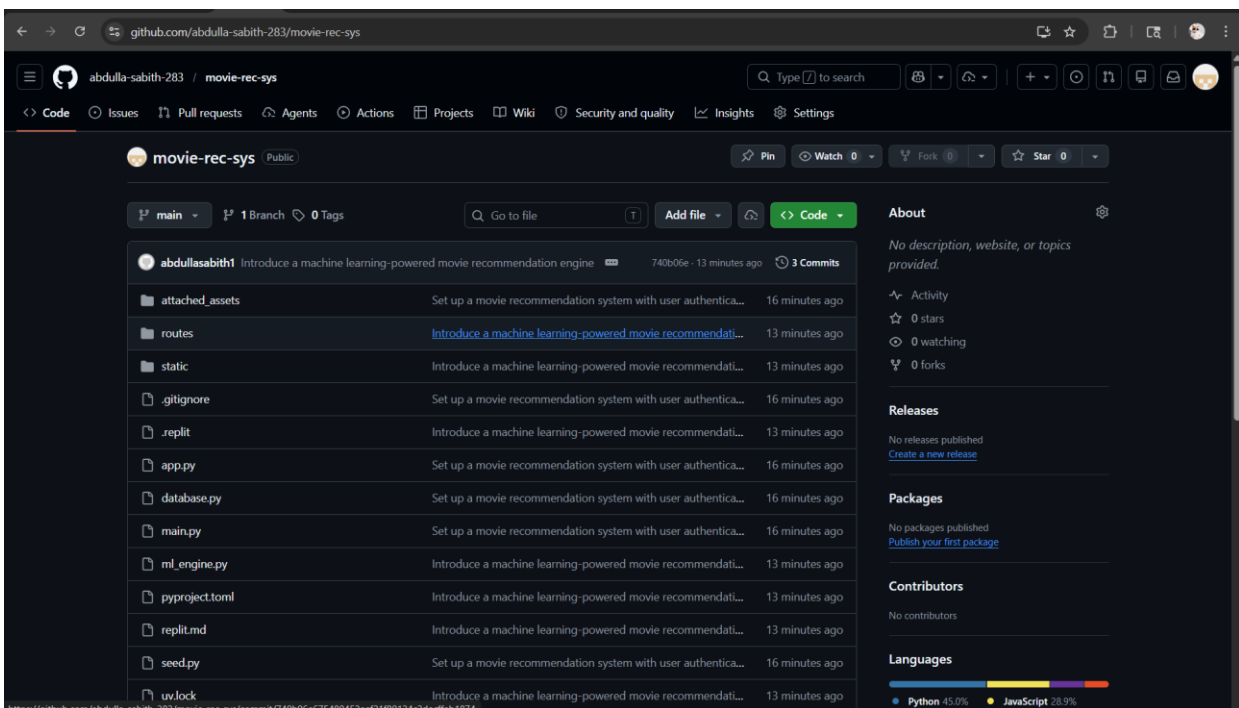
EXP NO: 10

## GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

### Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Music Playlist Batch Creator project.

### GitHub Project Structure



### Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.